DAY 23 | HANDS-ON ACTIVITIES

RAG, Embeddings and Vector DBs: Problem Statements

Seven problems on the handbook plus IT and travel FAQs. Offline embeddings are lexical stand-ins; add a key for real semantic embeddings.

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How to work: read the problem, write the steps in plain English first, then write the code. Test with the sample run, then try your own inputs, including wrong ones.

Problem 1: Similarity Matrix

Level: Warm-up | **Concepts:** embeddings, cosine

Embed 6 sentences (two about leave, two about travel, two about passwords) and print the cosine similarity matrix. Which pairs are most similar?

Sample run / expected result

```
leave-1 vs leave-2: 0.71 (online)
leave-1 vs password-1: 0.08
```

Hints

- With unit vectors, similarity is vecs @ vecs.T.

Problem 2: Flat Index

Level: Easy | **Concepts:** FAISS IndexFlatIP

Build a FAISS flat index over all chunks and return the top 3 chunks for five questions.

Sample run / expected result

```
Q: hotel limit in Japan -> c14, c13, c12
```

Hints

- faiss.IndexFlatIP(d); index.add(vecs.astype("float32"))

Problem 3: Exact vs Approximate

Level: Medium | **Concepts:** HNSW, recall, speed

On 20,000 random vectors, compare IndexFlatIP and IndexHNSWFlat: average query time and recall at 10.

Sample run / expected result

```
Flat: 3.1 ms/query, recall 1.00
HNSW: 0.2 ms/query, recall 0.96
```

Hints

- Recall = overlap of top-10 sets / 10.

Problem 4: BM25 for Codes

Level: Easy | **Concepts:** keyword search

Use BM25 to search for "VPN-417" and "14 characters". Compare the top result with vector search.

Sample run / expected result

```
BM25 top: c9 (VPN-417 ...)
Vector top (online): c9 or c10
```

Hints

- Tokenise to lowercase words, keep codes like vpn-417 together.

Problem 5: Hybrid with RRF

Level: Medium | **Concepts:** reciprocal rank fusion

Fuse BM25 and vector rankings with RRF for 8 test questions and compute hit rate at 3 for BM25, vector and hybrid.

Sample run / expected result

```
hit@3  bm25 0.75 | vector 0.75 | hybrid 0.88
```

Hints

- Write expected chunk ids for each question first.

Problem 6: Re-rank the Shortlist

Level: Challenging | **Concepts:** re-ranking

Retrieve the top 8 with hybrid search, re-rank with an LLM relevance score (online) or a word-overlap score (offline), and keep the top 3.

Sample run / expected result

```
Before re-rank: correct chunk at rank 4
After: rank 1
```

Hints

- Ask the model for a single integer 0 to 10 per candidate.

Problem 7: Grounded Answers with a Threshold

Level: Stretch | **Concepts:** grounded generation, refusal

If the best re-ranked score is below a threshold, answer "I don't know" without calling the model. Tune the threshold on answerable and unanswerable questions.

Sample run / expected result

```
Threshold 0.25: 7/8 answerable answered, 2/2 unanswerable refused
```

Hints

- Show both kinds of error: false refusals and false answers.

Ganbatte (頑張って): do your best